Literature Review & Project Context

Project Overview

Purpose

This thesis builds a pipeline that extracts motion capture data from existing 2D archival films and uses it to animate 3D character models in a VR environment. The primary goal is compelling, high-quality visual output — smooth, artifact-free VR character animation driven by real historical film performances. The thesis contribution is evaluated primarily on visual and experiential impact, with technical methodology in service of that goal.

End-to-End Pipeline (Current State)

The full pipeline is already working end-to-end:

MMPose (2D/3D pose estimation on film footage) → **JSON output** (per-frame, per-instance, per-joint keypoint data with confidence scores) → **Unity** (VR character animation)

This pipeline is complete. The thesis contribution sits between MMPose output and Unity: a **joint reliability estimator** that flags, downweights, or corrects untrustworthy joint predictions before they reach the VR animation system. Without this component, the animation exhibits visible jitter, phantom limbs, and anatomically implausible poses — artifacts that directly degrade the VR experience.

MMPose Configuration

- Model: MMPose COCO wholebody (133 joints)
- Output: per-frame JSON with keypoints [[x,y], ...] and keypoint_scores [...] per instance
- Instance tracking: DeepSORT-style track IDs
- Films processed: 9 films (Ramona + 8 others), primarily archival, black and white
- **Active joint set:** Although MMPose outputs 133 joints, the thesis reliability estimator and cleaning pipeline operates exclusively on the 17 standard COCO joints, as these are the only joints used in the downstream 3D data consumed by Unity. The 133-joint wholebody output is used only as a source for these 17 joints.

The 17 COCO joints used:

```
0:  nose              9:  left_wrist
1:  left_eye          10: right_wrist
2:  right_eye         11: left_hip
3:  left_ear          12: right_hip
4:  right_ear         13: left_knee
5:  left_shoulder     14: right_knee
6:  right_shoulder    15: left_ankle
7:  left_elbow        16: right_ankle
8:  right_elbow
```

Important implication: Current annotation coverage is primarily lower body joints (hips, knees, ankles). To have a complete reliability estimator for all 17 joints, annotation coverage needs to be extended to upper body joints (shoulders, elbows, wrists) and head/face joints.

The Core Technical Problem

MMPose was not designed for archival film. Two systematic failure modes dominate:

Failure Mode 1 — Overconfident scores: MMPose confidence scores are compressed near 1.0 (typically 0.88–0.99) regardless of whether a joint is visible, occluded, hallucinated, or geometrically implausible. Empirically demonstrated via beta distribution fitting: MLE drives parameters to degenerate spike distributions because score variance is near zero. The scalar confidence score is essentially uninformative as a reliability signal.

Failure Mode 2 — Always-present skeleton: MMPose always returns all joints whether or not they are visible in the frame. Occluded, off-screen, and hallucinated joints are returned with high confidence alongside genuinely visible joints, with no distinction in the output.

These two failure modes together mean that downstream animation systems have no reliable signal for which joints to trust, producing the visual artifacts that motivate this thesis.

Why Rule-Based Approaches Fail

Bone-length ratio filtering in 2D pixel space was evaluated as a baseline. Results: near-zero recall on human-labeled geometric implausibility cases. The fundamental reason is that 2D projection geometry in film is too variable — camera angle, partial visibility, character motion, and period costume all make fixed anthropometric bounds unreliable. Human visual judgment using temporal context (surrounding frames) correctly identifies trustworthy joints that rule-based filtering would wrongly reject, and vice versa. This negative result is a key thesis finding that motivates the learned approach.

Thesis Contribution

A learned per-joint reliability estimator trained on a novel hand-annotated dataset of film

footage, producing a trust signal that:

- Replaces MMPose's uninformative scalar confidence
- Enables downstream masking, downweighting, or interpolation of untrustworthy joints
- Directly improves VR animation quality — the primary evaluation criterion

The contribution is evaluated primarily through visual/experiential impact: before-and-after comparison of VR animation quality with and without the reliability filter, targeting compelling demonstration of artifact removal and motion smoothness.

Cleaned JSON Output Strategy

The thesis contribution outputs a modified version of the MMPose JSON where untrustworthy joint predictions have been corrected before Unity consumes them. The cleaning strategy is a two-pass temporal scheme:

Reliability categories and their treatment:

- `trust` — MMPose coordinate used directly. These are the anchor points for all interpolation.
- `partial_trust` — Blended between MMPose coordinate and a temporally interpolated estimate from surrounding trust frames. Blend weight is proportional to annotator confidence: higher confidence = more weight on MMPose coordinate. Suggested blend range: $\alpha \in [0.3, 0.7]$.
- `dont_trust` — MMPose coordinate ignored entirely. Joint position interpolated from nearest trust frames on either side temporally. If no trust frame exists within N frames, hold at last known trust position.
- `ambiguous` — Treated conservatively as partial_trust with low blend weight.

Two-pass scheme:

Forward pass: process frames sequentially, interpolating don't trust joints and blending partial trust joints from preceding trust frames.

Backward pass: for initialization, find the first frame containing a reliable whole-body detection and propagate it backwards to fill frames before it. A reliable whole-body detection requires at minimum: both hips, at least one knee, and root joint all labeled trust or high-confidence partial trust.

Joint subset hierarchy for initialization:

```
Core (required for initialization):
→ Both hips (11, 12), at least one knee (13 or 14), root

Extended (needed for upper body):
→ Both shoulders (5, 6), at least one elbow (7 or 8)

Full (ideal but not required):
→ Ankles (15, 16), wrists (9, 10), head (0-4)
```

Open question — Unity coordinate consumption: It is not yet confirmed whether Unity applies the 3D joint coordinates as direct world-space bone positions or converts them to bone rotations. This is a critical unknown: if Unity uses direct positions, coordinate-space interpolation is sufficient. If Unity converts to rotations, interpolating in position space may produce visual artifacts (unnatural joint bending, gimbal lock) that rotation-space interpolation would avoid. This needs to be confirmed before finalizing the cleaning strategy.

Output format: Modified JSON matching the MMPose output schema, with unreliable joint coordinates replaced by interpolated values. An additional field per joint flags whether the coordinate is original MMPose output or interpolated, preserving transparency for downstream debugging.

Key Empirical Findings So Far

- MMPose confidence scores carry almost no discriminative signal for occluded joints in film footage (demonstrated empirically via beta distribution fitting)
- Bone-length ratio filtering fails on film data — 126/128 geometric implausibility cases missed, near-zero recall (demonstrated via annotation comparison)
- Human temporal inference correctly trusts joints that geometric filtering rejects, and correctly distrusts joints that geometric filtering passes — rule-based approaches cannot replicate this
- Left knee is the most problematic joint in Ramona due to persistent skirt occlusion across hundreds of frames
- ~70% of joints in Ramona are occluded in some way
- Left-side joints are systematically harder to annotate and predict than right-side joints

Annotation Assets

Dataset 1 — Original occlusion annotation (~70k rows, Ramona + other films): Rich but complex annotation scheme with 5 visibility categories, multi-label occlusion reasons, severity, and annotator confidence. Not the primary training data. Used for exploratory analysis and

motivating findings. Has known quality limitations from single-annotator inconsistency and scheme complexity.

Dataset 2 — New reliability annotation (~2,000 rows, in progress): Purpose-built for the thesis contribution. See Dataset Strategy section below for full schema.

Paper Summaries & Relevance

---

1. Gonçalves, Jesus & Jorge (2024) — *Motion Capture in Mixed-Reality Applications: A Deep Denoising Approach*

Venue: Virtual Worlds 2024

Summary: Addresses the problem of correcting occluded and displaced joint predictions from a Microsoft Kinect sensor for use in mixed-reality applications. A ResNet-based deep neural network is trained on CMU mocap data that has been synthetically corrupted with two noise types: occlusions (joints displaced to origin) and displacements (small Gaussian perturbations). The CMU skeleton is adapted to match the Kinect 17-joint structure. A Savitzky-Golay temporal filter is applied post-network to reduce jitter. Results are evaluated on CMU test data (MAE) and on Kinect data via a 12-person perceptual study since no ground truth exists for real Kinect captures.

Key finding: The model trained exclusively with occlusion noise achieves the lowest MAE on CMU test data. For Kinect data, the model trained with both noise types produces the most visually natural results. Adding a small number of real Kinect poses to training data reduces error by 6.3%.

Relevance to this thesis: This is the most structurally analogous paper in the set. Their core problem — a pose detector produces noisy/occluded joint predictions that must be cleaned before driving a 3D character — directly mirrors the thesis problem. However, a critical distinction exists: the Kinect explicitly sets confidence to zero for occluded joints, providing a clean binary occlusion signal. MMPose provides no such signal — it remains highly confident on occluded joints. This thesis solves a strictly harder problem.

Extractable methods:

- Savitzky-Golay filter as a lightweight temporal smoother post-reliability estimation
- Synthetic noise corruption framework: CMU mocap data can be corrupted to simulate MMPose-style occlusion errors and used to train a pose correction network
- ResNet denoising architecture as a baseline or component

- Perceptual evaluation methodology (user study comparing animation quality) as a downstream evaluation strategy when ground truth 3D is unavailable

---

2. Yu, Wang & Li (2025) — *Animation Pose Generation Model Based on Kinect Depth Image and Occlusion-Robust Pose-Maps Algorithm*

Venue: IEEE Access 2025

Summary: Proposes a dynamic pose generation model combining Kinect depth images with monocular visual pose estimation. Uses spatiotemporal sparse sampling to construct 3D skeleton sequences, matrix completion to repair missing depth data, dynamic median filtering for motion smoothing, and a Kalman filter data fusion strategy to combine depth-based and RGB-based pose estimates. Evaluated on Human3.6M and MPI-INF-3DHP, achieving 87.7% and 86.8% correct limb percentage respectively.

Relevance to this thesis: Like Gonçalves, this paper has access to depth information that this thesis does not. However, their Kalman filter fusion strategy is directly relevant — specifically the idea of weighting temporal updates by per-joint confidence. In this thesis, the reliability score from the learned estimator could serve as the Kalman update weight: high reliability joints update strongly from the current frame, low reliability joints rely more on the temporal prediction from previous frames. This provides a principled, confidence-weighted temporal smoothing mechanism that is more adaptive than fixed-weight filters.

Extractable methods:

- Kalman filter with confidence-weighted updates as a temporal smoothing strategy
- Matrix completion / missing data imputation techniques for handling untrustworthy joint predictions
- Evaluation metrics: correct limb percentage (CLP), percentage of correct keypoints (PCK), percentage of detected joints (PDJ) — relevant benchmarks for downstream 3D pose quality

---

3. Luvizon, Habermann, Golyanik, Kortylewski & Theobalt (2024) — *Scene-Aware 3D Multi-Human Motion Capture from a Single Camera*

Venue: MPI Informatics

Summary: The most technically sophisticated paper in this set. Proposes a non-linear optimization framework that jointly estimates absolute 3D position, articulated pose, body shape, and scene scale for multiple humans from a single static RGB camera. Leverages large-

scale pretrained models for 2D joints, joint angles, normalized disparity maps, and segmentation masks. Reconstructs a 3D point cloud of the static scene and performs space-time coherent optimization over the full video to enforce temporal, spatial, and physical plausibility. Outperforms previous methods on multi-person 3D pose benchmarks and generalizes to in-the-wild conditions.

Relevance to this thesis: This paper is the closest in domain to the thesis — it works from RGB video without depth sensors, handles multiple people, and addresses temporal consistency. Their use of normalized disparity maps for depth estimation connects directly to the question of whether monocular depth models (ZoeDepth, Depth Anything) could be used in the thesis pipeline to provide a depth signal for improving reliability estimation and 3D lifting. Their space-time coherent optimization over the full video is a sophisticated version of the temporal smoothing problem this thesis addresses.

Extractable methods:

- Normalized disparity estimation as a proxy depth signal — worth testing on archival film footage to assess feasibility
- Space-time coherent optimization framework as a conceptual basis for temporal consistency enforcement
- Joint optimization over pose, shape, and scene scale as a long-term direction for the 3D lifting component
- Their treatment of in-the-wild generalization is relevant to the cross-movie generalization challenge in this thesis

Important caveat: Their footage is modern RGB video with known camera parameters. Archival film introduces additional challenges (grain, low resolution, black and white, unknown camera geometry) that may limit direct applicability of their depth estimation approach.

---

4. Gargi, Kim, Liang & Ma — *Monocular Pose Estimation for VR Motion Capture Applications*

Venue: University of Michigan course project

Summary: A course-level project proposing a convolutional network pipeline for 2D and 3D pose estimation from a single webcam for VR motion capture. Uses a VGG-19 encoder with a cascade convolutional decoder for 2D heatmap prediction, followed by a separate network for z-coordinate prediction. Trained on CMU Panoptic Studio data. Evaluates using MPJPE (mean per joint position error).

Relevance to this thesis: Limited technical depth but relevant as context. Their framing of VR motion capture from a single camera as a monocular 2D-to-3D lifting problem is essentially the

same downstream task this thesis feeds into. The MPJPE metric is a standard evaluation measure for 3D pose quality that could be used to evaluate the downstream impact of this thesis's reliability estimator — better masking of untrustworthy joints should produce lower MPJPE on the 3D lifted output.

Extractable methods:

- MPJPE as a downstream evaluation metric
- 2D-to-3D lifting architecture as a reference for the downstream component of the pipeline
- Their handling of incomplete skeletons (filtering images where not all joints are present) as a reference point for how existing work handles missing joints — by exclusion rather than reliability weighting

---

5. Elhayek, Kovalenko, Murthy, Malik & Stricker — *Fully Automatic Multi-person Human Motion Capture for VR Applications*

Venue: DFKI / University of Kaiserslautern

Summary: Proposes a fully automatic real-time multi-person motion capture system using a monocular RGB camera. Automatically estimates the number of persons, generates person-specific 3D skeletons, and estimates initial 3D locations. For each frame, fits 3D skeletons to 2D body-part detections from a CNN-based 2D pose estimator by maximizing an objective function combining a skeleton fitting term with motion and pose priors. Handles dynamic entry and exit of persons. Demonstrates on outdoor scenes, community videos, and mobile phone footage.

Relevance to this thesis: Their skeleton fitting energy function — combining a data fitting term with motion priors and pose priors — is a principled formulation that is directly relevant to the thesis pipeline. Specifically, their motion prior (joints cannot jump instantaneously) and pose prior (skeleton can only take anatomically valid poses) are the same constraints this thesis needs to enforce when propagating joint predictions across frames. Their work on community videos and mobile phone footage (lower quality, uncontrolled conditions) is the closest in spirit to working with archival film, though still far from the challenges of 1920s-1940s cinema.

Extractable methods:

- Skeleton fitting energy function combining data term + motion prior + pose prior as a framework for temporal consistency
- Automatic person detection and skeleton initialization as a reference for the instance tracking component upstream
- Their treatment of per-frame jitter (which they note as a limitation of offline methods) motivates the temporal smoothing component of this thesis

Dataset Strategy: From Occlusion Detection to Reliability Estimation

The Original 70k Annotation Dataset

The original annotation effort produced approximately 70,000 rows of richly labeled data across 9 films, with per-joint labels for visibility category (1–5), occlusion reason (multi-label), occlusion severity, annotator confidence, and reason for low confidence. This dataset was built around the question: "what is happening to this joint visually?"

This dataset is not the primary training data for the thesis contribution. While it provides valuable exploratory signal and motivated the core empirical findings (MMPose confidence is uninformative; geometric filtering fails on film data), it has several limitations that make it unsuitable as the primary training set:

- Single annotator for most films, introducing consistency concerns
- Annotation scheme is complex and subjective, particularly for boundary cases between visibility categories
- The richness of the scheme (5 visibility categories, multi-label reasons, severity) adds noise without necessarily adding the signal needed for the downstream task
- The question it answers ("what is the occlusion state of this joint") is not exactly the question the downstream model needs answered

The New Reliability Dataset

The thesis contribution is trained on a new, purpose-built dataset that reframes the annotation task around a single, well-defined question:

"Should the downstream model trust MMPose's prediction for this joint?"

This is a reliability estimation problem, not an occlusion detection problem. The distinction is important: a joint can be occluded and still have a trustworthy MMPose prediction (if surrounding frames provide strong temporal evidence), and a joint can be fully visible but have an untrustworthy prediction (if MMPose has placed it in a geometrically implausible location). Occlusion is one cause of unreliability but not the only one, and it does not automatically imply the prediction should be discarded.

New Dataset Schema

Each row in the new dataset corresponds to one MMPose joint prediction and contains the following features:

Identity features (from MMPose output):

- `frame_id` — frame index in the video
- `instance_id` — which character instance (0, 1, ...)
- `track_id` — persistent track ID across frames
- `joint_id` — numeric joint identifier (COCO wholebody)
- `joint_name` — human-readable joint name

MMPose prediction features:

- `x`, `y` — predicted joint coordinates in pixel space
- `mmpose_confidence` — MMPose's scalar confidence score (typically 0.88–0.99)

Computed geometric features (derived automatically, not annotated):

- `dist_to_boundary` — Euclidean distance from predicted joint to nearest frame edge; joints near the boundary are more likely to be off-screen
- `geom_plausible` — boolean result of bone-length ratio check against pelvis width reference (True/False/None if reference unavailable)
- `geom_flag` — reason for geometric implausibility if flagged (e.g. ratio outside bounds, zero-length bone, no pelvis reference)
- `bone_length` — pixel-space Euclidean distance to parent joint
- `bone_ratio` — bone length normalized by pelvis width

Reliability annotation (human-labeled, the target signal):

- `reliability_category` — one of: `trust`, `partial_trust`, `dont_trust`, `ambiguous`
- `reason_for_distrust` — one of: `unverifiable_occlusion`, `clearly_wrong`, `geometric_implausibility`, `hallucinated`, `poor_image_quality`, `outside_frame` (null if trust or ambiguous)
- `annotator_confidence` — 3-level scale: certain / fairly sure / unsure

Validity:

- `valid` — boolean flag for whether the skeleton instance is a real person (screens hallucinated skeletons)

Key Design Decisions

The annotation task is presented to the annotator as a viewer showing the full frame with a bounding box centered on MMPose's predicted joint location, plus ±2 surrounding frames for temporal context. This design means:

- The annotator is explicitly evaluating MMPose's prediction, not making an independent ground truth localization
- Temporal context is available, allowing Trust judgments on occluded joints when surrounding frames confirm the prediction
- The bounding box anchors the annotator to MMPose's coordinate, keeping the annotation task aligned with the downstream use case

Relationship to Occlusion Detection

This dataset is related to but distinct from occlusion detection. Prior work on occlusion-aware pose estimation (e.g. Cheng et al., Song et al.) treats occlusion as the primary signal for filtering or correcting joints. This thesis treats reliability as the primary signal, with occlusion being one of several contributing factors. The reasons for distrust in this dataset capture a broader set of failure modes than occlusion alone — geometric implausibility, hallucination, and frame boundary effects are equally important in the film domain and are not captured by occlusion-focused approaches.

This framing also means the dataset generalizes more naturally across films and domains: the reliability question is answerable for any pose detector output, regardless of whether the failure mode is occlusion-specific.

Key Distinctions: This Thesis vs. Existing Work

Aspect	Existing papers	This thesis
Source material	Live camera / controlled capture	Archival 2D film
Confidence signal	Clean (Kinect=0 for occlusion) or absent	Systematically overconfident, uninformative
Occlusion cause	Body self-occlusion, sensor limits	Costume, staging, camera angle, film quality
Camera control	Known parameters, static or calibrated	Unknown, cinematic, unconstrained

Aspect	Existing papers	This thesis
Depth information	Kinect depth or estimated from modern RGB	Unknown; monocular depth untested on archival film
Annotation data	Standard benchmarks (H36M, CMU)	Novel hand-annotated film dataset with occlusion semantics
Application	Live VR interaction	Retargeting historical performance to VR character

The fundamental contribution of this thesis is not a new motion capture system — it is a **reliability estimation framework for 2D pose detector output in the archival film domain**, where no existing confidence signal is trustworthy and occlusion patterns are driven by cinematic rather than sensor constraints.